

Real-World Problems Faced by Farmers

AgriNovaAI

Introduction

Agriculture is one of the most important sectors in the world. It provides food, raw materials, and employment for billions of people. In many developing countries, especially in Asia, agriculture is the main source of income for millions of families. Countries such as Sri Lanka, India, Bangladesh, Nepal, Pakistan, Vietnam and Thailand depend heavily on agriculture for food production, economic growth and rural development. Farmers play a vital role in ensuring food security and supporting the national economy.

However, modern farming is becoming more difficult every year. Farmers face many challenges before planting crops, during crop growth and even after harvesting. These challenges reduce crop production, increase farming costs, lower farmers' income and threaten food security. Many of these problems are becoming more serious because of climate change, population growth, environmental degradation and changing market conditions.

In Sri Lanka, most farmers are smallholder farmers who cultivate relatively small areas of land. They often depend on traditional farming knowledge, personal experience and seasonal weather patterns when making important farming decisions. However, unpredictable weather, limited access to modern technology, rising production costs, labour shortages, pest and disease outbreaks, water shortages, poor market access and limited financial resources make farming increasingly difficult. Similar problems are experienced by farmers across many Asian countries.

Every day, farmers make many important decisions that directly affect their crops and income. They must decide when to prepare the land, when to plant seeds, how much fertilizer to apply, when to irrigate, how to control pests and diseases, when to harvest and where to sell their products. Many of these decisions are made with limited information. In many cases, farmers rely on personal experience, advice from neighbouring farmers or local traders because they do not have access to accurate, real-time information. As a result, incorrect decisions can lead to reduced crop yields, financial losses, wasted resources and environmental damage.

Although governments, agricultural extension officers, research institutions and private organizations provide various support services, many farmers still struggle to receive timely, accurate and location-specific advice. Agricultural experts cannot visit every farm regularly, especially in remote rural areas. In addition, many existing agricultural information systems provide only general recommendations and are unable to consider the farmer's specific crop, location, weather conditions, soil characteristics and farming objectives. This creates a significant gap between available agricultural knowledge and the practical decisions farmers need to make every day.

At the same time, agriculture is entering a new era of digital transformation. Technologies such as Artificial Intelligence (AI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, Internet of Things (IoT) sensors, satellite imagery, drones, weather forecasting systems and Geographic Information Systems (GIS) are creating new opportunities to improve agricultural decision-making. These technologies can help farmers identify crop diseases, monitor weather conditions, manage irrigation, optimize fertilizer use, predict market prices and access reliable agricultural knowledge more quickly than ever before.

However, developing an effective AI-powered agricultural assistant requires a deep understanding of the real-world problems faced by farmers. Before designing intelligent systems, it is essential to understand how farmers work, what decisions they make, what challenges they face, why existing solutions are often insufficient, and where technology can provide practical support. Without understanding these real-world farming problems, it is difficult to build AI systems that truly meet farmers' needs.

Therefore, this report provides a comprehensive analysis of the major real-world challenges faced by farmers, with particular attention to Sri Lanka and other Asian countries. It examines the complete farming lifecycle, from planning and land preparation to harvesting and marketing and explains the technical, environmental, economic and social issues that farmers encounter at each stage. The report also analyses the root causes of these challenges, their impacts on farmers and food production, the limitations of current solutions and the opportunities for applying modern AI technologies to address these problems.

The knowledge presented in this report serves as a foundation for developing **AgriNovaAI**, an Agentic AI-powered agricultural assistant that combines Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), intelligent AI agents and trusted agricultural knowledge to provide reliable, explainable and context-aware farming support. By understanding farmers' real needs and daily challenges, AgriNovaAI aims to help farmers make better decisions, improve agricultural productivity, reduce risks and contribute to sustainable agriculture and food security in Sri Lanka and across Asia.

Aim

The aim of this report is to identify and analyse the real-world problems faced by farmers, especially in Sri Lanka and other Asian countries.

Objectives

The main objectives of this report are:

- To identify the major real-world problems faced by farmers.
- To analyse the root causes and impacts of these problems.
- To understand the challenges farmers face during different stages of farming.
- To evaluate the limitations of current farming methods and existing solutions.

Scope

This report focuses on the major challenges faced by farmers in Sri Lanka and other Asian countries. It covers environmental, agricultural, economic, social and technological issues that affect farming. The report analyses these problems, their causes and impacts.

Real-World Problems Faced by Farmers

Problem 1: Land Degradation - The Ground Beneath Farming Is Wearing Out

Overview

Land degradation is one of the biggest challenges facing agriculture today. Healthy soil is the foundation of farming because it provides nutrients, water and support for crop growth. However, continuous farming without proper soil management gradually reduces soil quality. As a result, the land becomes less fertile and crop production decreases.

Land degradation is becoming a serious problem in many countries, including Sri Lanka, India, Bangladesh and other parts of Asia. It threatens food production, farmer income and environmental sustainability.

What is Land Degradation?

Land degradation is the process in which agricultural land loses its natural ability to produce healthy crops. Over time, the soil becomes less productive because its nutrients, organic matter and structure are damaged or removed.

Healthy soil contains minerals, microorganisms, organic matter, air and water. These components help plants grow properly. When the soil is degraded, it cannot provide enough nutrients or retain enough water, making farming more difficult.

Why Does Land Degradation Happen?

Several factors contribute to land degradation.

- Continuous cultivation of the same crop without crop rotation.
- Excessive use of chemical fertilizers and pesticides.
- Soil erosion caused by heavy rainfall and strong winds.
- Deforestation and removal of natural vegetation.
- Poor irrigation practices that increase soil salinity.
- Overgrazing by livestock.
- Climate change and extreme weather conditions.

Many small-scale farmers continue to use traditional farming methods without giving the soil enough time to recover.

Real-World Example

A rice farmer in Sri Lanka has cultivated paddy on the same field for many years. To maintain production, the farmer applies larger amounts of chemical fertilizer every season. Although the crops initially grow well, the soil gradually loses its organic matter and beneficial microorganisms.

After several years, the farmer notices that crop yields are decreasing even though more fertilizer is being applied. During heavy rains, the topsoil is washed away, making the problem even worse. The farmer now spends more money while producing fewer crops.

Impact on Farmers

Land degradation creates many challenges for farmers.

- Lower crop yields.
- Increased fertilizer costs.
- Poor water retention.
- Reduced soil fertility.
- Higher production costs.
- Greater risk of crop failure.

- Lower farm income.

For smallholder farmers, these problems can directly affect their family's income and food security.

Current Solutions

Farmers and agricultural organizations use several methods to reduce land degradation.

- Crop rotation.
- Organic compost and manure.
- Cover crops.
- Conservation tillage.
- Tree planting.
- Soil testing.
- Balanced fertilizer application.

These methods improve soil fertility and reduce erosion.

Limitations of Current Solutions

Although these methods are effective, many farmers cannot fully implement them because of:

- Limited knowledge.
- High costs.
- Lack of technical support.
- Limited access to soil testing services.
- Short-term focus on immediate crop production.

AI Opportunities

Artificial Intelligence can help farmers monitor soil health and improve land management.

Possible AI applications include:

- Soil health analysis.
- Fertilizer recommendations.
- Crop suitability prediction.
- Satellite-based land monitoring.
- Early detection of soil degradation.

An AI assistant can provide personalized recommendations based on soil type, crop type, weather and farming history.

Summary

Land degradation is a slow but serious agricultural problem that reduces soil fertility and crop productivity. Proper soil management and modern technologies can help restore soil health and improve long-term agricultural sustainability.

Problem 2: Water Scarcity - Running Out of the One Thing Farming Cannot Do Without

Overview

Water is one of the most important resources in agriculture. Every crop needs sufficient water to grow properly. However, many regions around the world are experiencing water shortages due to climate change, increasing population, poor water management and excessive water use.

Water scarcity is becoming a major challenge for farmers in Sri Lanka and many Asian countries.

What is Water Scarcity?

Water scarcity occurs when the available water is not enough to meet agricultural, domestic and industrial needs.

Farmers depend on rainfall, rivers, reservoirs, wells and irrigation systems. When these water sources become limited, crop production is affected.

Why Does Water Scarcity Happen?

Several factors contribute to water shortages.

- Low rainfall.
- Long drought periods.
- Climate change.
- Overuse of groundwater.
- Poor irrigation systems.
- Water pollution.
- Increasing demand for water from growing populations.

In many farming areas, irrigation systems are old and inefficient, causing significant water loss.

Real-World Example

A vegetable farmer in the dry zone of Sri Lanka depends on reservoir water for irrigation. During a prolonged drought, water levels become very low. The farmer must reduce irrigation, causing vegetables to grow slowly and produce lower yields.

The farmer also spends more money pumping groundwater, increasing production costs.

Impact on Farmers

Water scarcity affects almost every farming activity.

- Reduced crop growth.
- Lower yields.
- Poor crop quality.
- Higher irrigation costs.
- Increased crop failures.
- Reduced farmer income.

Smallholder farmers are often the most affected because they have limited access to alternative water sources.

Current Solutions

Several methods are used to improve water management.

- Drip irrigation.
- Rainwater harvesting.
- Reservoir construction.
- Efficient irrigation scheduling.
- Water conservation practices.
- Drought-resistant crop varieties.

These methods improve water use efficiency.

Limitations of Current Solutions

Many farmers still face difficulties because:

- Irrigation systems are expensive.
- Limited technical knowledge.
- Poor infrastructure.
- Lack of real-time weather information.
- Limited financial support.

AI Opportunities

Artificial Intelligence can improve irrigation management by using:

- Weather forecasts.
- Soil moisture sensors.
- Satellite imagery.
- Crop growth information.
- Water usage analysis.

AI systems can recommend the best time and amount of irrigation, reduce water waste while improve crop growth.

Summary

Water scarcity is one of the greatest threats to agriculture. Efficient water management, combined with AI-based decision support, can help farmers use available water more effectively and improve agricultural productivity.

Problem 3: Climate Change - Heat, Drought, and Unpredictability Are Cutting Into Yields

Overview

Climate change is one of the most serious challenges affecting modern agriculture. Rising temperatures, changing rainfall patterns, floods, droughts and extreme weather events are making farming more difficult every year.

Farmers can no longer rely on traditional seasonal weather patterns because the climate has become increasingly unpredictable.

What is Climate Change?

Climate change refers to long-term changes in temperature, rainfall and weather conditions caused mainly by increasing greenhouse gas emissions.

These changes directly affect agriculture because crops depend heavily on stable weather conditions.

Why Does Climate Change Affect Farming?

Climate change affects farming in several ways.

- Higher temperatures reduce crop growth.
- Drought reduces water availability.
- Heavy rainfall causes flooding and soil erosion.
- Strong winds damage crops.
- New pests and diseases spread more easily.
- Flowering and harvesting seasons become less predictable.

As a result, farming becomes more risky and less profitable.

Real-World Example

A rice farmer in Sri Lanka expects rainfall during the planting season. However, the rains arrive several weeks late. After planting, an unexpected drought damages the young rice plants.

Later in the season, heavy rainfall floods the field just before harvest, causing additional crop losses.

The farmer experiences both drought and flooding within the same season.

Impact on Farmers

Climate change creates many challenges.

- Lower crop yields.
- Increased crop failures.
- More pest and disease outbreaks.
- Higher production costs.
- Greater financial uncertainty.
- Reduced food security.
- Increased environmental damage.

These problems particularly affect small-scale farmers who have limited resources.

Current Solutions

Farmers are adopting various adaptation strategies.

- Climate-resilient crop varieties.
- Improved irrigation.
- Crop diversification.
- Weather forecasting.
- Soil conservation.
- Water management.

Governments and research organizations also provide climate adaptation programs.

Limitations of Current Solutions

Despite these efforts, several challenges remain.

- Weather forecasts may not always be accurate at the local level.
- Climate conditions change rapidly.
- Modern technologies are expensive.
- Many farmers lack training.
- Limited access to climate information.

AI Opportunities

Artificial Intelligence can support climate-smart agriculture by providing:

- Weather forecasting.
- Climate risk prediction.
- Crop suitability analysis.
- Irrigation recommendations.
- Early warning systems.
- Disaster risk alerts.

AI assistants can combine weather data, satellite information and crop knowledge to help farmers make better daily decisions.

Summary

Climate change is transforming agriculture around the world. Farmers need accurate information, better planning and intelligent decision-support systems to adapt to changing weather conditions. AI technologies can play an important role in helping farmers reduce risks, improve productivity and strengthen food security.

Problem 4: Pests, Diseases, and Invasive Species - A Moving Target

Overview

Pests, plant diseases and invasive species are among the biggest challenges in modern agriculture. Every year, farmers lose a large portion of their crops because insects, fungi, bacteria, viruses and invasive plants damage crops before they can be harvested. Climate change and global trade are causing these problems to spread more quickly, making crop protection more difficult than ever before.

What are Pests, Diseases and Invasive Species?

Pests are insects or animals that damage crops by feeding on leaves, stems, roots, fruits or seeds. Examples include caterpillars, aphids and stem borers.

Plant diseases are caused by microorganisms such as fungi, bacteria, viruses and nematodes. They reduce plant growth, damage leaves and fruits and can destroy entire fields if not controlled.

Invasive species are plants, insects or animals that spread into new areas where they do not naturally belong. They often grow rapidly because they have few natural enemies and compete with crops for nutrients, water and sunlight.

Why Does This Problem Happen?

Several factors increase pest and disease outbreaks.

- Rising temperatures help insects reproduce faster.
- Climate change allows pests to spread into new regions.
- Heavy rainfall creates favourable conditions for fungal diseases.
- Monocropping increases pest populations.
- International trade introduces new invasive species.
- Incorrect pesticide use leads to pesticide resistance.

As a result, farmers must constantly deal with new pests and diseases that were not common in the past.

Real-World Example

A rice farmer in Sri Lanka notices that the leaves of his paddy field are turning yellow and drying. He is unsure whether the problem is caused by insects, a fungal disease or nutrient deficiency. Without expert advice, he buys a pesticide recommended by a local shop. Unfortunately, the pesticide is not suitable for the actual problem and the disease continues to spread throughout the field. By harvest time, the farmer loses a significant portion of his crop and spends additional money on ineffective treatments.

Impact on Farmers

Pests and diseases create many challenges.

- Reduced crop yield.
- Lower crop quality.
- Increased pesticide costs.

- More frequent spraying.
- Higher production costs.
- Financial losses.
- Reduced food production.

Some farmers may even lose their entire harvest if outbreaks are not controlled quickly.

Current Solutions

Farmers use different methods to manage pests and diseases.

- Regular field inspection.
- Chemical pesticides.
- Biological pest control.
- Disease-resistant crop varieties.
- Crop rotation.
- Integrated Pest Management (IPM).

These methods help reduce crop damage but require proper knowledge and timing.

Limitations of Current Solutions

Current methods still have several limitations.

- Farmers may misidentify pests or diseases.
- Expert advice is not always available.
- Excessive pesticide use can harm the environment.
- Some pests become resistant to pesticides.
- Monitoring large farms is difficult.

AI Opportunities

Artificial Intelligence can improve pest and disease management by using:

- Computer Vision to identify pests and diseases from crop images.
- Machine Learning to predict disease outbreaks.
- Weather data to estimate disease risk.
- RAG systems to retrieve reliable treatment recommendations.
- AI assistants to provide instant advice.

Summary

Pests, diseases and invasive species continue to threaten global food production. Early detection, accurate diagnosis and timely management are essential. AI-powered systems can help farmers identify problems quickly, reduce unnecessary pesticide use and protect crop production.

Problem 5: Pollinator Decline - A Quiet Threat to Nutrition, Not Just Yield

Overview

Pollinators such as bees, butterflies, birds and other insects play a vital role in agriculture. They transfer pollen between flowers, allowing many crops to produce fruits, vegetables, seeds and nuts. However, pollinator populations are declining around the world, creating a serious but often overlooked agricultural problem.

What is Pollinator Decline?

Pollinator decline refers to the reduction in the number and diversity of pollinating insects and animals.

Many important food crops depend on pollinators for successful reproduction. Without enough pollinators, crop yields decrease and food quality is affected.

Why Does This Problem Happen?

Several factors contribute to pollinator decline.

- Habitat destruction.
- Excessive pesticide use.
- Climate change.
- Air pollution.
- Diseases affecting bee populations.
- Loss of flowering plants.

These factors reduce pollinator populations and limit their ability to support agriculture.

Real-World Example

A fruit farmer notices that many flowers bloom during the season, but only a small number develop into fruits. After investigation, agricultural officers find that the number of bees in the area has decreased because nearby natural habitats were removed and pesticides were used frequently. As a result, pollination was insufficient, reducing fruit production.

Impact on Farmers

Pollinator decline affects agriculture in several ways.

- Lower fruit production.
- Reduced vegetable yields.
- Smaller crop sizes.
- Poor fruit quality.
- Lower farmer income.
- Reduced food diversity.

This problem also affects consumers because food prices increase when production decreases.

Current Solutions

Current conservation methods include:

- Protecting natural habitats.
- Planting flowering plants.
- Reducing harmful pesticide use.
- Supporting beekeeping.
- Promoting biodiversity.

These practices help maintain healthy pollinator populations.

Limitations of Current Solutions

Several challenges remain.

- Farmers often focus only on crop production.
- Habitat restoration takes time.
- Climate change continues to affect pollinators.
- Limited public awareness.

AI Opportunities

Artificial Intelligence can support pollinator conservation by:

- Monitoring pollinator populations.
- Identifying pollinator-friendly farming practices.
- Predicting pollination risks.
- Recommending environmentally friendly pest control methods.

AI can help balance crop protection with biodiversity conservation.

Summary

Pollinator decline is a hidden agricultural problem that affects food production, nutrition, and biodiversity. Protecting pollinators is essential for sustainable agriculture and future food security.

Problem 6: Labor Shortages and an Aging Farming Workforce

Overview

Agriculture depends heavily on human labour. However, many countries are experiencing a shortage of agricultural workers because fewer young people are interested in farming. At the same time, many experienced farmers are growing older, creating a workforce imbalance that threatens future agricultural production.

What is Labor Shortage?

Labor shortage occurs when there are not enough workers to perform farming activities such as planting, irrigation, harvesting and transportation.

An aging farming workforce means that the average age of farmers continues to increase while fewer young people enter the agricultural sector.

Why Does This Problem Happen?

Several factors contribute to labour shortages.

- Young people migrate to cities.
- Farming is physically demanding.
- Agricultural income is often unstable.
- Better job opportunities exist in other industries.
- Population aging.

As older farmers retire, there are fewer replacements.

Real-World Example

A vegetable farmer in Sri Lanka plans to harvest tomatoes during the peak season. However, there are not enough workers available because many young people have moved to urban areas for employment. The harvest is delayed and many tomatoes become overripe before reaching the market, reducing both quality and income.

Impact on Farmers

Labor shortages create many challenges.

- Delayed planting.
- Delayed harvesting.
- Higher labour costs.
- Lower crop quality.
- Reduced productivity.
- Increased financial pressure.

Smallholder farmers often struggle the most because they cannot afford expensive machinery.

Current Solutions

Current approaches include:

- Agricultural machinery.
- Mechanized harvesting.
- Community labour sharing.
- Training programs.
- Government support.

These methods improve efficiency but are not always affordable.

Limitations of Current Solutions

Several challenges remain.

- Machinery is expensive.
- Small farms are difficult to mechanize.
- Skilled machine operators are limited.
- Rural populations continue to decline.

AI Opportunities

Artificial Intelligence can reduce labour requirements through:

- Smart farming systems.
- Autonomous agricultural machinery.
- Crop monitoring drones.
- AI-based farm planning.
- Automated irrigation.
- Farm management assistants.

These technologies improve efficiency while reducing dependence on manual labour.

Summary

Labor shortages are becoming a serious threat to agricultural production. Modern technologies and intelligent automation can help farmers improve productivity while reducing labour requirements.

Problem 7: Financial Exclusion and Post-Harvest Losses - Losing Food and Money Before It Reaches the Market

Overview

Many farmers successfully produce crops but still struggle to earn a good income. Limited access to financial services and significant post-harvest losses prevents farmers from receiving the full value of their hard work. These problems are especially common among smallholder farmers in developing countries.

What is Financial Exclusion and Post-Harvest Loss?

Financial exclusion means that farmers cannot easily access banking services, loans, insurance or agricultural credit.

Post-harvest loss refers to the loss of crops after harvesting because of poor storage, transportation, handling, pests or spoilage.

Why Does This Problem Happen?

Several factors contribute to these challenges.

- Limited access to agricultural loans.
- High interest rates.
- Poor storage facilities.

- Inadequate transportation.
- Lack of cold storage.
- Poor market access.
- Limited financial literacy.

These factors reduce farmers' profits even after successful crop production.

Real-World Example

A mango farmer produces a high-quality harvest but has no refrigerated storage facility. The nearest wholesale market is several hours away. During transportation, many fruits become damaged and spoil before they can be sold. At the same time, the farmer cannot obtain a low-interest loan to invest in better storage equipment. As a result, the farmer earns much less than expected despite a successful harvest.

Impact on Farmers

These problems create serious financial challenges.

- Reduced profits.
- Food waste.
- Increased debt.
- Limited investment.
- Lower production efficiency.
- Reduced food availability.

These losses also affect national food security and increase consumer prices.

Current Solutions

Current solutions include:

- Agricultural loans.
- Crop insurance.
- Warehouse facilities.
- Cold storage.
- Farmer cooperatives.
- Better transportation networks.

These services improve market access and reduce crop losses.

Limitations of Current Solutions

Many farmers still face difficulties because:

- Financial services are limited in rural areas.
- Loan approval is difficult.
- Storage facilities are expensive.

- Poor infrastructure delays transportation.
- Small farmers have limited bargaining power.

AI Opportunities

Artificial Intelligence can support farmers by providing:

- Market price prediction.
- Financial planning.
- Harvest scheduling.
- Supply chain optimization.
- Storage management.
- Smart logistics planning.
- Personalized financial recommendations.

AI can also help farmers decide when to harvest, where to sell and how to maximize profits while minimizing post-harvest losses.

Summary

Financial exclusion and post-harvest losses prevent many farmers from achieving stable incomes. Improving financial access, storage infrastructure and market connections supported by AI-based decision systems can significantly increase farmer profitability while reducing food waste and strengthening food security.

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